

UPPER BOUND ON THE PRODUCT OF THE NUMBER OF ELEMENTS IN TWO SETS WITH UNIQUE PRODUCTS

CHATGPT

ABSTRACT. This is a blueprint written by ChatGPT in order to let Aristotle formalize a lower bound on the largest possible value of $|A||B|$ for two sets $A, B \subseteq \{1, 2, \dots, n\}$ such that all products ab with $a \in A, b \in B$ are distinct. I first let ChatGPT write up Szemerédi's 1976 proof that shows that

$$|A||B| \leq \frac{cn^2}{\log n}$$

for some absolute constant c and all $n > 1$, and then asked it to optimize the constant c as much as possible. This eventually led to showing that $c = 54.43$ is admissible, if n is large enough. This paper is entirely AI-generated, not independently verified, and not optimized for readability or ease of understanding. That being said, the proof is (essentially) correct, as the Lean file that Aristotle produced for an upper bound with $c = 60$ can be found here:

<https://github.com/Woett/Lean-files/blob/main/ErdosProblem490.lean>

All logarithms are natural. The notation $o(1)$ means $n \rightarrow \infty$, uniformly in the finite sets and indices occurring in the corresponding estimate.

Definition 1. *Let*

$$\mathbb{N}_{\geq 1} = \{m \in \mathbb{N} : 1 \leq m\}, \quad \mathbb{N}_{\geq 0} = \{m \in \mathbb{N} : 0 \leq m\}, \quad \mathcal{P} = \{p \in \mathbb{N}_{\geq 1} : p \text{ is prime}\}.$$

For $X \in \mathbb{R}$, *put*

$$[X] = \{m \in \mathbb{N}_{\geq 1} : m \leq X\}.$$

If $S \subseteq \mathbb{N}_{\geq 1}$ *is finite and* $p \in \mathcal{P}$, *put*

$$S[p] = \{s \in S : p \mid s\}, \quad p^{-1}S[p] = \{t \in \mathbb{N}_{\geq 1} : pt \in S\}.$$

Definition 2. *Let* $n \in \mathbb{N}_{\geq 1}$. *A pair* (A, B) *is* n -*admissible if*

$$A, B \subseteq [n]$$

and the map

$$A \times B \rightarrow \mathbb{N}_{\geq 1}, \quad (a, b) \mapsto ab$$

is injective.

Definition 3. *For* $\lambda > 1$ *and* $k \in \mathbb{N}_{\geq 0}$, *put*

$$Y_{\lambda, k} = 2\lambda^k, \quad I_{\lambda, k} = \{p \in \mathcal{P} : Y_{\lambda, k} \leq p < Y_{\lambda, k+1}\},$$

$$N_{\lambda, k} = |I_{\lambda, k}|, \quad M_{\lambda, k} = \prod_{p \leq Y_{\lambda, k+1}} \left(1 - \frac{1}{p}\right).$$

If $0 \leq r \leq N_{\lambda, k}$, *put*

$$E_{\lambda, k}(r) = \max_{\substack{T \subseteq I_{\lambda, k} \\ |T| \leq r}} \prod_{p \in T} \left(1 - \frac{1}{p}\right)^{-1}.$$

If $m : \mathbb{N}_{\geq 0} \rightarrow \mathbb{N}_{\geq 0}$ satisfies $m_k \leq N_{\lambda,k}$ for every k , define

$$D_{\lambda,m} = \prod_{k=0}^{\infty} E_{\lambda,k}(m_k).$$

If moreover $g : \mathbb{N}_{\geq 0} \rightarrow [1, \infty)$, define

$$\Omega_{\lambda,m,g} = \sum_{\substack{k \in \mathbb{N}_{\geq 0} \\ m_k < N_{\lambda,k}}} \frac{g_k N_{\lambda,k}}{Y_{\lambda,k} M_{\lambda,k} \sqrt{m_k + 1}}.$$

The triple (λ, m, g) is admissible if

$$D_{\lambda,m} < \infty, \quad \Omega_{\lambda,m,g} < 1, \quad g_k \rightarrow \infty.$$

Definition 4. Let (λ, m, g) be admissible. For $p \in I_{\lambda,k}$, put

$$w_{\lambda,m,g}(p) = \begin{cases} \frac{g_k}{Y_{\lambda,k} M_{\lambda,k} \sqrt{m_k + 1}}, & m_k < N_{\lambda,k}, \\ 0, & m_k = N_{\lambda,k}. \end{cases}$$

A finite set $S \subseteq \mathbb{N}_{\geq 1}$ is (λ, m, g) -regular if, for every $p \in \mathcal{P}$,

$$w_{\lambda,m,g}(p) > 0 \text{ and } S[p] \neq \emptyset \implies |S[p]| > w_{\lambda,m,g}(p) |S|.$$

Definition 5. For $x \in \mathbb{R}$, put

$$\pi(x) = |\{p \in \mathcal{P} : p \leq x\}|$$

and

$$\psi(x) = \sum_{\substack{p \in \mathcal{P}, a \in \mathbb{N}_{\geq 1} \\ p^a \leq x}} \log p.$$

Empty sums and empty products have values 0 and 1, respectively.

Theorem 6 (External input: Dusart). *The following three estimates are used as external input.*

(1) For every $x \geq 2278382$,

$$\left| \prod_{p \leq x} \left(1 - \frac{1}{p}\right) - \frac{1}{e^\gamma \log x} \right| \leq \frac{1}{5e^\gamma \log^4 x}.$$

This is Dusart [1, Theorem 5.9].

(2) For every $x \geq 88789$,

$$\frac{x}{\log x} + \frac{x}{\log^2 x} + \frac{2x}{\log^3 x} \leq \pi(x) \leq \frac{x}{\log x} + \frac{x}{\log^2 x} + \frac{2.53816x}{\log^3 x}.$$

This is Dusart [1, Corollary 5.2].

(3) For every $x \geq 2$,

$$|\psi(x) - x| < \frac{1.66x}{\log^2 x}.$$

This is Dusart [1, Theorem 3.3].

Lemma 7 (Mertens product from Dusart). *As $x \rightarrow \infty$,*

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma} + o(1)}{\log x}.$$

Proof. For $x \geq 2278382$, Theorem 6(1) gives

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) = \frac{e^{-\gamma}}{\log x} + O\left(\frac{1}{\log^4 x}\right).$$

Since $\log^{-4} x = o(\log^{-1} x)$, the assertion follows. $\square$

Definition 8. Let $f : \mathbb{N}_{\geq 1} \rightarrow \{0, 1\}$. For $X \geq 1$, put

$$F_f(X) = \sum_{m \leq X} f(m), \quad H_f(X) = \sum_{m \leq X} \frac{f(m)}{m},$$

and

$$L_f(X) = \sum_{m \leq X} f(m) \log m.$$

The function f is completely multiplicative if

$$f(ab) = f(a)f(b) \quad (a, b \in \mathbb{N}_{\geq 1}).$$

Lemma 9 (Logarithmic convolution bound). Let $f : \mathbb{N}_{\geq 1} \rightarrow \{0, 1\}$ be completely multiplicative. Then, for every $X \geq 1$,

$$L_f(X) \leq \sum_{a \leq X} f(a) \psi(X/a).$$

Proof. For $m \in \mathbb{N}_{\geq 1}$,

$$\log m = \sum_{d|m} \Lambda(d),$$

where $\Lambda(d) = \log p$ if $d = p^a$ with $p \in \mathcal{P}$ and $a \in \mathbb{N}_{\geq 1}$, and $\Lambda(d) = 0$ otherwise. Hence

$$\begin{aligned} L_f(X) &= \sum_{m \leq X} f(m) \sum_{d|m} \Lambda(d) \\ &= \sum_{a \leq X} \sum_{d \leq X/a} f(ad) \Lambda(d) \\ &= \sum_{a \leq X} f(a) \sum_{d \leq X/a} f(d) \Lambda(d) \\ &\leq \sum_{a \leq X} f(a) \sum_{d \leq X/a} \Lambda(d) = \sum_{a \leq X} f(a) \psi(X/a). \end{aligned}$$

$\square$

Lemma 10 (Block estimate). Let $A > \log 2$. Put

$$C_A = \psi(e^A), \quad \delta_A = \frac{1.66}{A^2}.$$

Let $f : \mathbb{N}_{\geq 1} \rightarrow \{0, 1\}$ be completely multiplicative. If $X > e^{A+C_A}$, then

$$F_f(X) - F_f(Xe^{-A}) \leq \frac{(1 + \delta_A)X H_f(X)}{\log X - A - C_A}.$$

Proof. Put

$$D = F_f(X) - F_f(Xe^{-A}).$$

Every integer counted by D is larger than Xe^{-A} , and therefore

$$D(\log X - A) \leq L_f(X).$$

By Lemma 9,

$$L_f(X) \leq \sum_{a \leq Xe^{-A}} f(a)\psi(X/a) + \sum_{Xe^{-A} < a \leq X} f(a)\psi(X/a).$$

If $a \leq Xe^{-A}$, then $X/a \geq e^A \geq 2$, and Theorem 6(3) gives

$$\psi(X/a) \leq \left(1 + \frac{1.66}{\log^2(X/a)}\right) \frac{X}{a} \leq (1 + \delta_A) \frac{X}{a}.$$

If $Xe^{-A} < a \leq X$, then $1 \leq X/a < e^A$, so $\psi(X/a) \leq C_A$. Therefore

$$L_f(X) \leq (1 + \delta_A)XH_f(X) + C_AD.$$

Combining this with the lower bound for $L_f(X)$ gives the assertion. $\square$

Lemma 11 (Mean estimate from Dusart). *Uniformly for completely multiplicative functions $f : \mathbb{N}_{\geq 1} \rightarrow \{0, 1\}$,*

$$F_f(X) \leq (1 + o(1)) \frac{X}{\log X} H_f(X) \quad (X \rightarrow \infty).$$

Proof. Fix $A > \log 2$, and let C_A and δ_A be as in Lemma 10. For large X , put

$$J = \left\lceil \frac{2 \log \log X}{A} \right\rceil, \quad X_j = Xe^{-jA} \quad (0 \leq j \leq J).$$

Then $e^{AJ} \geq (\log X)^2$, and hence

$$F_f(X_J) \leq X_J H_f(X_J) \leq \frac{XH_f(X)}{(\log X)^2}.$$

For $0 \leq j < J$, Lemma 10 applied to X_j gives

$$F_f(X_j) - F_f(X_{j+1}) \leq \frac{(1 + \delta_A)X_j H_f(X_j)}{\log X_j - A - C_A}.$$

Since $H_f(X_j) \leq H_f(X)$ and

$$\log X_j - A - C_A = \log X - (j+1)A - C_A = \log X + O_A(\log \log X)$$

uniformly for $0 \leq j < J$, summing the preceding inequalities gives

$$\begin{aligned} F_f(X) &\leq \frac{XH_f(X)}{(\log X)^2} + \frac{(1 + \delta_A + o_A(1))XH_f(X)}{\log X} \sum_{j=0}^{J-1} e^{-jA} \\ &\leq \left(\frac{1 + \delta_A}{1 - e^{-A}} + o_A(1) \right) \frac{XH_f(X)}{\log X}. \end{aligned}$$

Taking $\limsup_{X \rightarrow \infty}$, and then letting $A \rightarrow \infty$, proves the claim. $\square$

Lemma 12 (Asymptotic upper-bound sieve from Dusart). *Let $X = X(n) \rightarrow \infty$, let $P = P(n) \subseteq \mathcal{P} \cap [X]$ be finite, and put*

$$Q(X, P) = \{m \in [X] : p \nmid m \text{ for every } p \in P\}.$$

Then

$$|Q(X, P)| \leq (e^\gamma + o(1))X \prod_{p \in P} \left(1 - \frac{1}{p}\right),$$

where the $o(1)$ is uniform in $P \subseteq \mathcal{P} \cap [X]$.

Proof. Define $f : \mathbb{N}_{\geq 1} \rightarrow \{0, 1\}$ by

$$f(m) = 1 \iff \text{no prime divisor of } m \text{ belongs to } P.$$

Then f is completely multiplicative and

$$|Q(X, P)| = F_f(X).$$

Let

$$R = (\mathcal{P} \cap [X]) \setminus P.$$

Since every prime divisor of an integer $m \leq X$ lies in $\mathcal{P} \cap [X]$,

$$H_f(X) \leq \prod_{p \in R} \left(1 - \frac{1}{p}\right)^{-1}.$$

Lemma 11 gives

$$|Q(X, P)| \leq (1 + o(1)) \frac{X}{\log X} \prod_{p \in R} \left(1 - \frac{1}{p}\right)^{-1}.$$

Finally,

$$\begin{aligned} \frac{1}{\log X} \prod_{p \in R} \left(1 - \frac{1}{p}\right)^{-1} &= \frac{\prod_{p \in P} \left(1 - \frac{1}{p}\right)}{\log X \prod_{p \leq X} \left(1 - \frac{1}{p}\right)} \\ &= (e^\gamma + o(1)) \prod_{p \in P} \left(1 - \frac{1}{p}\right) \end{aligned}$$

by Lemma 7. This proves the assertion. $\square$

Lemma 13. *Let $S \subseteq \mathbb{N}_{\geq 1}$ be finite and let $p \in \mathcal{P}$. The map*

$$S[p] \rightarrow p^{-1}S[p], \quad s \mapsto s/p,$$

is a bijection. In particular,

$$|p^{-1}S[p]| = |S[p]|.$$

Proof. The inverse map is $t \mapsto pt$. $\square$

Lemma 14 (Weighted deletion). *Let (λ, m, g) be admissible, and let $S \subseteq \mathbb{N}_{\geq 1}$ be finite and nonempty. Then there exists an (λ, m, g) -regular subset $S^* \subseteq S$ such that*

$$|S^*| > (1 - \Omega_{\lambda, m, g}) |S|.$$

Proof. Put $w = w_{\lambda, m, g}$ and $\Omega = \Omega_{\lambda, m, g}$. Define $S^0 = S$. If S^i has been defined, put

$$R_i = \{p \in \mathcal{P} : w(p) > 0, \ S^i[p] \neq \emptyset, \ |S^i[p]| \leq w(p) |S^i|\},$$

and

$$S^{i+1} = S^i \setminus \bigcup_{p \in R_i} S^i[p].$$

If $p \in R_i$, then $S^{i+1}[p] = \emptyset$, so each prime belongs to at most one R_i . Let S^* be the first terminal set and put $R = \bigcup_i R_i$. Then

$$\begin{aligned} |S \setminus S^*| &\leq \sum_i \sum_{p \in R_i} |S^i[p]| \\ &\leq |S| \sum_{p \in R} w(p) \\ &\leq \Omega |S|. \end{aligned}$$

The terminal condition is exactly regularity. $\square$

Lemma 15 (Weighted regular reduction). *Let (λ, m, g) be admissible, and let (A, B) be n -admissible with $A, B \neq \emptyset$. Then there are (λ, m, g) -regular subsets $A^* \subseteq A$, $B^* \subseteq B$ such that (A^*, B^*) is n -admissible and*

$$|A^*| |B^*| > (1 - \Omega_{\lambda, m, g})^2 |A| |B|.$$

Proof. Apply Lemma 14 to A and to B . Admissibility is inherited by passing to subsets. $\square$

Lemma 16 (Collision lemma). *Let (A, B) be n -admissible. Let $p, q \in \mathcal{P}$, $p \neq q$. If*

$$p^{-1}A[p] \cap q^{-1}A[q] \neq \emptyset,$$

then

$$p^{-1}B[p] \cap q^{-1}B[q] = \emptyset.$$

Proof. Choose $x \in p^{-1}A[p] \cap q^{-1}A[q]$. If $y \in p^{-1}B[p] \cap q^{-1}B[q]$, then

$$(px)(qy) = (qx)(py),$$

with $px, qx \in A$ and $py, qy \in B$. Admissibility gives $px = qx$, hence $p = q$, contradiction. $\square$

Definition 17. For $\lambda > 1$, finite $A, B \subseteq \mathbb{N}_{\geq 1}$, and $k \in \mathbb{N}_{\geq 0}$, put

$$L_{\lambda, k}(A, B) = \{p \in I_{\lambda, k} : A[p] \neq \emptyset, B[p] \neq \emptyset\}.$$

Lemma 18 (Euler product over common prime divisors). *Let $\lambda > 1$, let $m : \mathbb{N}_{\geq 0} \rightarrow \mathbb{N}_{\geq 0}$ satisfy $m_k \leq N_{\lambda, k}$ for every k , and let $A, B \subseteq [n]$. If*

$$|L_{\lambda, k}(A, B)| \leq m_k \quad (k \in \mathbb{N}_{\geq 0}),$$

then

$$\prod_{\substack{p \leq n \\ A[p] \neq \emptyset \\ B[p] \neq \emptyset}} \left(1 - \frac{1}{p}\right)^{-1} \leq D_{\lambda, m}.$$

Proof. In each interval $I_{\lambda, k}$, the common primes form a subset of cardinality at most m_k . The corresponding partial product is at most $E_{\lambda, k}(m_k)$. Multiplication over k gives the claim. $\square$

Definition 19. For $n \geq 2$, $\lambda > 1$, $k \in \mathbb{N}_{\geq 0}$, and $S \subseteq [n]$, put

$$P_S(n, \lambda, k) = \{p \in \mathcal{P} : Y_{\lambda, k+1} < p \leq n/Y_{\lambda, k}, S[p] = \emptyset\}$$

and

$$\Pi_S(n, \lambda, k) = \prod_{p \in P_S(n, \lambda, k)} \left(1 - \frac{1}{p}\right).$$

Lemma 20 (Asymptotic sifted bounds). *Let $\lambda > 1$, let $S \subseteq [n]$, and let $k \in \mathbb{N}_{\geq 0}$. Then*

$$|S| \leq (e^\gamma + o(1))n\Pi_S(n, \lambda, k).$$

Moreover, for every

$$L \subseteq \{p \in I_{\lambda, k} : S[p] \neq \emptyset\},$$

one has

$$\left| \bigcup_{p \in L} p^{-1}S[p] \right| \leq (e^\gamma + o(1)) \frac{n}{Y_{\lambda, k}} \Pi_S(n, \lambda, k).$$

Proof. The first estimate follows from Lemma 12, with $X = n$ and $P = P_S(n, \lambda, k)$.

For the second estimate, put

$$U = \bigcup_{p \in L} p^{-1}S[p].$$

Then $U \subseteq [n/Y_{\lambda, k}]$. If $r \in P_S(n, \lambda, k)$, $x \in U$, and $px \in S$ with $p \in L$, then $p < Y_{\lambda, k+1} < r$. Hence $r \mid x$ would imply $r \mid px$, contrary to $S[r] = \emptyset$. Thus $U[r] = \emptyset$ for all $r \in P_S(n, \lambda, k)$. If $P_S(n, \lambda, k) \neq \emptyset$, then $n/Y_{\lambda, k} > Y_{\lambda, k+1}$, so $n/Y_{\lambda, k} \rightarrow \infty$ uniformly in this case, and Lemma 12 applies with $X = n/Y_{\lambda, k}$. If $P_S(n, \lambda, k) = \emptyset$, then the trivial bound $|U| \leq n/Y_{\lambda, k}$ is stronger. $\square$

Lemma 21 (Weighted interval product). *Let $\lambda > 1$, and let $g : \mathbb{N}_{\geq 0} \rightarrow [1, \infty)$ satisfy $g_k \rightarrow \infty$. Uniformly for $k \in \mathbb{N}_{\geq 0}$,*

$$\frac{M_{\lambda, k}}{g_k} \prod_{\substack{p \in \mathcal{P} \\ Y_{\lambda, k+1} < p \leq n/Y_{\lambda, k}}} \left(1 - \frac{1}{p}\right) \leq \frac{e^{-\gamma} + o(1)}{\log n}.$$

Proof. Put

$$Q(n, k) = \frac{M_{\lambda, k}}{g_k} \prod_{Y_{\lambda, k+1} < p \leq n/Y_{\lambda, k}} \left(1 - \frac{1}{p}\right).$$

We use Theorem 6(1) directly, rather than an asymptotic Mertens statement. In particular, uniformly for $x \geq 2278382$,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) \leq \frac{e^{-\gamma}}{\log x} + O\left(\frac{1}{\log^4 x}\right).$$

First suppose that $Y_{\lambda, k+1} \geq n/Y_{\lambda, k}$. Then the product in the definition of $Q(n, k)$ is empty and $Y_{\lambda, k} \geq (n/\lambda)^{1/2}$. Therefore

$$M_{\lambda, k} \leq \frac{e^{-\gamma} + o(1)}{\log Y_{\lambda, k+1}} \leq \frac{2e^{-\gamma} + o(1)}{\log n},$$

uniformly in this range. Since the least such k tends to infinity with n , and since $g_k \rightarrow \infty$, this gives

$$Q(n, k) \leq \frac{o(1)}{\log n}$$

uniformly in the empty-product range.

Now suppose that $Y_{\lambda, k+1} < n/Y_{\lambda, k}$. Then

$$Q(n, k) = \frac{1}{g_k} \prod_{p \leq n/Y_{\lambda, k}} \left(1 - \frac{1}{p}\right),$$

and also $Y_{\lambda,k} < (n/\lambda)^{1/2}$, so $n/Y_{\lambda,k} > (\lambda n)^{1/2}$. Fix $K \in \mathbb{N}_{\geq 0}$. For $k \leq K$, the explicit Dusart bound gives

$$Q(n, k) \leq \frac{e^{-\gamma} + o(1)}{\log(n/Y_{\lambda,k})} = \frac{e^{-\gamma} + o(1)}{\log n},$$

uniformly over this finite set of k . For $k > K$, the same explicit bound and $n/Y_{\lambda,k} > (\lambda n)^{1/2}$ give

$$Q(n, k) \leq \frac{2e^{-\gamma} + o(1)}{g_k \log(n/Y_{\lambda,k})} \leq \frac{4e^{-\gamma} + o(1)}{g_k \log n}.$$

Since $\inf_{k > K} g_k \rightarrow \infty$ as $K \rightarrow \infty$, the contribution of $k > K$ is $o(1)/\log n$. Let $K \rightarrow \infty$. $\square$

Lemma 22 (High interval product). *Let $\lambda > 1$, let $m : \mathbb{N}_{\geq 0} \rightarrow \mathbb{N}_{\geq 0}$ satisfy $m_j \leq N_{\lambda,j}$ for every j , and let $A, B \subseteq [n]$. Suppose*

$$|L_{\lambda,j}(A, B)| \leq m_j \quad (j > k).$$

Then

$$\prod_{\substack{p \in \mathcal{P} \\ Y_{\lambda,k+1} < p \leq n/Y_{\lambda,k} \\ A[p] \neq \emptyset \\ B[p] \neq \emptyset}} \left(1 - \frac{1}{p}\right)^{-1} \leq D_{\lambda,m}.$$

Proof. Every prime in the product lies in an interval $I_{\lambda,j}$ with $j > k$. Applying the definition of $E_{\lambda,j}(m_j)$ interval by interval and then multiplying gives a bound by $\prod_{j > k} E_{\lambda,j}(m_j) \leq D_{\lambda,m}$. $\square$

Lemma 23 (Multiplicity lemma). *Let I be finite and let $\{E_i\}_{i \in I}$ be finite sets. Put*

$$U = \bigcup_{i \in I} E_i.$$

If

$$\sum_{i \in I} |E_i| > M |U|,$$

then there exists $x \in U$ such that

$$|\{i \in I : x \in E_i\}| > M.$$

Proof. Sum the multiplicity function on U . $\square$

Lemma 24 (Weighted small alternative). *Let (λ, m, g) be admissible, let $\varepsilon > 0$, and let (A, B) be n -admissible. Let $k \in \mathbb{N}_{\geq 0}$, and suppose*

$$|L_{\lambda,j}(A, B)| \leq m_j \quad (j > k).$$

If either

$$|A| < (e^\gamma + \varepsilon) \frac{M_{\lambda,k}}{g_k} n \Pi_A(n, \lambda, k)$$

or

$$|B| < (e^\gamma + \varepsilon) \frac{M_{\lambda,k}}{g_k} n \Pi_B(n, \lambda, k),$$

then

$$|A| |B| \leq (e^\gamma + \varepsilon + o(1)) D_{\lambda,m} \frac{n^2}{\log n}.$$

Proof. By symmetry assume the first alternative. Lemma 20 gives

$$|B| \leq (e^\gamma + o(1))n\Pi_B(n, \lambda, k).$$

Thus

$$|A| |B| \leq (e^{2\gamma} + e^\gamma \varepsilon + o(1)) \frac{M_{\lambda,k}}{g_k} n^2 \Pi_A(n, \lambda, k) \Pi_B(n, \lambda, k).$$

For every prime p in the interval $Y_{\lambda,k+1} < p \leq n/Y_{\lambda,k}$,

$$\mathbf{1}_{A[p]=\emptyset} + \mathbf{1}_{B[p]=\emptyset} \geq 1 - \mathbf{1}_{A[p] \neq \emptyset, B[p] \neq \emptyset}.$$

Hence $\Pi_A \Pi_B$ is at most the full interval product times the reciprocal product over common primes. Lemma 22 bounds the reciprocal product by $D_{\lambda,m}$, and Lemma 21 bounds the full interval product. $\square$

Lemma 25 (Weighted large interval case). *Let (λ, m, g) be admissible, let $\varepsilon > 0$, and let (A, B) be n -admissible. Assume that A and B are (λ, m, g) -regular. Suppose there is $k \in \mathbb{N}_{\geq 0}$ such that*

$$|L_{\lambda,k}(A, B)| > m_k$$

and

$$|L_{\lambda,j}(A, B)| \leq m_j \quad (j > k).$$

Then

$$|A| |B| \leq (e^\gamma + \varepsilon + o(1)) D_{\lambda,m} \frac{n^2}{\log n}.$$

Proof. Let $L = L_{\lambda,k}(A, B)$. Since $|L| > m_k$, one has $m_k < N_{\lambda,k}$. Put $r = m_k + 1$ and

$$c_k = \frac{g_k}{Y_{\lambda,k} M_{\lambda,k} \sqrt{r}}.$$

If one of the alternatives in Lemma 24 holds, the result follows. Assume neither holds. Thus

$$|A| \geq (e^\gamma + \varepsilon) \frac{M_{\lambda,k}}{g_k} n \Pi_A(n, \lambda, k),$$

and the analogous inequality holds for B .

For $p \in L$, regularity and Lemma 13 give

$$|p^{-1}A[p]| = |A[p]| > c_k |A|.$$

Since $|L| \geq r$,

$$\sum_{p \in L} |p^{-1}A[p]| > (e^\gamma + \varepsilon) \sqrt{r} \frac{n}{Y_{\lambda,k}} \Pi_A(n, \lambda, k).$$

By Lemma 20, the union of the sets $p^{-1}A[p]$, $p \in L$, has cardinality at most

$$(e^\gamma + o(1)) \frac{n}{Y_{\lambda,k}} \Pi_A(n, \lambda, k).$$

Therefore, for all sufficiently large n , Lemma 23 gives an $x \in \mathbb{N}_{\geq 1}$ such that

$$L^* = \{p \in L : x \in p^{-1}A[p]\}$$

satisfies

$$|L^*| > \sqrt{r}.$$

For $p \in L^*$, regularity of B and Lemma 13 give

$$|p^{-1}B[p]| > c_k |B|.$$

Consequently,

$$\sum_{p \in L^*} |p^{-1}B[p]| > (e^\gamma + \varepsilon) \frac{n}{Y_{\lambda,k}} \Pi_B(n, \lambda, k).$$

By Lemma 20, this is larger than the union cardinality for all sufficiently large n . Hence some $y \in \mathbb{N}_{\geq 1}$ lies in $p^{-1}B[p] \cap q^{-1}B[q]$ for two distinct $p, q \in L^*$. Since $p, q \in L^*$, the same two primes also satisfy

$$x \in p^{-1}A[p] \cap q^{-1}A[q],$$

contradicting Lemma 16. Therefore one of the alternatives in Lemma 24 holds. $\square$

Lemma 26 (Small interval case). *Let $\lambda > 1$, let $m : \mathbb{N}_{\geq 0} \rightarrow \mathbb{N}_{\geq 0}$ satisfy $m_k \leq N_{\lambda,k}$ for every k , and let (A, B) be n -admissible. Assume*

$$|L_{\lambda,k}(A, B)| \leq m_k \quad (k \in \mathbb{N}_{\geq 0}).$$

Then

$$|A| |B| \leq (e^\gamma + o(1)) D_{\lambda,m} \frac{n^2}{\log n}.$$

Proof. Put

$$P_A = \{p \leq n : A[p] = \emptyset\}, \quad P_B = \{p \leq n : B[p] = \emptyset\}.$$

Lemma 12 gives

$$|A| \leq (e^\gamma + o(1)) n \prod_{p \in P_A} \left(1 - \frac{1}{p}\right),$$

and similarly for B . Therefore

$$|A| |B| \leq (e^{2\gamma} + o(1)) n^2 \prod_{p \leq n} \left(1 - \frac{1}{p}\right) \prod_{\substack{p \leq n \\ A[p] \neq \emptyset \\ B[p] \neq \emptyset}} \left(1 - \frac{1}{p}\right)^{-1}.$$

Now use Lemma 7 and Lemma 18. $\square$

Theorem 27 (Layer-weighted asymptotic bound). *Let (λ, m, g) be admissible. Then*

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \frac{\log n}{n^2} \sup_{\substack{A, B \subseteq [n] \\ (A, B) \text{ } n\text{-admissible}}} |A| |B| \\ & \leq \frac{e^\gamma D_{\lambda,m}}{(1 - \Omega_{\lambda,m,g})^2}. \end{aligned}$$

Proof. Fix $\varepsilon > 0$, and let (A, B) be n -admissible. If $A = \emptyset$ or $B = \emptyset$, there is nothing to prove. Otherwise apply Lemma 15 and rename the resulting regular subsets as A, B . This costs the factor $(1 - \Omega_{\lambda,m,g})^{-2}$.

If

$$|L_{\lambda,k}(A, B)| \leq m_k \quad (k \in \mathbb{N}_{\geq 0}),$$

then Lemma 26 gives the required bound. Otherwise choose the largest k such that $|L_{\lambda,k}(A, B)| > m_k$. Lemma 25 gives

$$|A| |B| \leq (e^\gamma + \varepsilon + o(1)) D_{\lambda,m} \frac{n^2}{\log n}.$$

Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$. $\square$

Definition 28. Put

$$\lambda_0 = 1.931 = \frac{1931}{1000}, \quad \mu_0 = 1.48035 = \frac{29607}{20000}.$$

For $k \in \mathbb{N}_{\geq 0}$, put

$$Y_k = Y_{\lambda_0, k}, \quad I_k = I_{\lambda_0, k}, \quad N_k = N_{\lambda_0, k}, \quad M_k = M_{\lambda_0, k}.$$

For $0 \leq r \leq N_k$, put

$$s_k(r) = \begin{cases} \frac{N_k}{Y_k M_k \sqrt{r+1}}, & r < N_k, \\ 0, & r = N_k, \end{cases} \quad \ell_k(r) = \log E_{\lambda_0, k}(r).$$

Let

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
b_k	2	2	2	3	7	11	19	30	56	96	176	304	564	994	1811	3293	5997

k	17	18	19	20	21	22	23	24
b_k	10966	20101	36919	17569	26601	40428	61648	94242

k	25	26	27	28	29	30	31	32
b_k	144399	221714	340923	525002	809293	1248806	1928495	2980115

and define

$$m_k = \begin{cases} b_k, & 0 \leq k \leq 32, \\ \lfloor \mu_0 Y_k^{2/3} \rfloor, & k \geq 33. \end{cases}$$

Finally, define

$$g_k = \begin{cases} 1, & k \leq 150, \\ (k - 149)^2, & k > 150. \end{cases}$$

Lemma 29 (Explicit prime-counting and Mertens consequences). *Let*

$$\pi_U(x) = \frac{x}{\log x} + \frac{x}{\log^2 x} + \frac{2.53816x}{\log^3 x}$$

and

$$\pi_L(x) = \frac{x}{\log x} + \frac{x}{\log^2 x} + \frac{2x}{\log^3 x}.$$

For every $x \geq 88789$,

$$\pi_L(x) \leq \pi(x) \leq \pi_U(x).$$

For every $x \geq 2278382$,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right) \geq \frac{e^{-\gamma}}{\log x} \left(1 - \frac{0.2}{(\log x)^3}\right).$$

Proof. The prime-counting inequalities are exactly Theorem 6(2). The product inequality is the lower half of Theorem 6(1). $\square$

Lemma 30 (Finite numerical certificate). *For the sequence in Definition 28,*

$$b_k \leq N_k \quad (0 \leq k \leq 32),$$

$$\sum_{k=0}^{32} \ell_k(b_k) < 3.277487, \quad \sum_{k=0}^{32} s_k(b_k) < 0.062611.$$

Proof. This is a finite exact enumeration of the primes in the intervals I_k , $0 \leq k \leq 32$. Since $(1 - 1/p)^{-1}$ is decreasing as p increases, the maximum in $E_{\lambda_0, k}(b_k)$ is obtained by taking the b_k smallest primes in I_k . The enumeration gives

$$\sum_{k=0}^{32} \ell_k(b_k) = 3.277486595644029 \dots < 3.277487$$

and, since $g_k = 1$ for $k \leq 32$,

$$\sum_{k=0}^{32} s_k(b_k) = 0.062610837348652 \dots < 0.062611.$$

It also verifies $b_k \leq N_k$ for $0 \leq k \leq 32$. □

Lemma 31 (Tail cardinalities). *For every $k \geq 33$,*

$$m_k < N_k.$$

Proof. For $k \geq 33$, interval arithmetic using Lemma 29 gives

$$\pi_L(Y_{k+1}) - \pi_U(Y_k) > 70Y_k^{2/3}.$$

The left side is a lower bound for N_k . Since $70 > \mu_0$,

$$N_k > \mu_0 Y_k^{2/3} > \lfloor \mu_0 Y_k^{2/3} \rfloor = m_k.$$

□

Lemma 32 (Tail comparison). *For every $k \geq 33$,*

$$\frac{N_k}{Y_k M_k} < 1.682294.$$

Proof. Since $Y_{33} > 88789$ and $Y_{34} > 2278382$, Lemma 29 gives, for $k \geq 33$,

$$N_k \leq \pi_U(Y_{k+1}) - \pi_L(Y_k)$$

and

$$M_k \geq \frac{e^{-\gamma}}{\log Y_{k+1}} \left(1 - \frac{0.2}{(\log Y_{k+1})^3} \right).$$

Thus it is enough to verify

$$\frac{\pi_U(Y_{k+1}) - \pi_L(Y_k)}{Y_k} \cdot \frac{e^\gamma \log Y_{k+1}}{1 - 0.2/(\log Y_{k+1})^3} < 1.682294.$$

The left side is decreasing for $k \geq 33$, and interval arithmetic gives its value at $k = 33$ as

$$1.682293385011672 \dots < 1.682294.$$

□

Lemma 33 (Tail sums). *For the sequence in Definition 28,*

$$\sum_{k=33}^{\infty} \frac{m_k}{Y_k - 1} < 0.004286$$

and

$$\sum_{k=33}^{\infty} g_k Y_k^{-1/3} < 0.002895224.$$

Proof. Put $\rho = \lambda_0^{-1/3}$. Since $Y_k = 2\lambda_0^k$, $Y_k^{-1/3} = 2^{-1/3}\rho^k$. Also $m_k \leq \mu_0 Y_k^{2/3}$, and hence

$$\frac{m_k}{Y_k - 1} \leq \frac{\mu_0 Y_k^{-1/3}}{1 - Y_k^{-1}}.$$

For $k \geq 33$, $Y_k \geq Y_{33}$, and direct interval evaluation gives

$$\frac{\mu_0}{1 - Y_{33}^{-1}} \sum_{k=33}^{\infty} Y_k^{-1/3} < 0.004286.$$

This proves the first inequality. The second follows from the finite geometric sum from $k = 33$ to 150, together with the closed forms

$$\sum_{j=1}^{\infty} j\rho^j = \frac{\rho}{(1-\rho)^2}, \quad \sum_{j=1}^{\infty} j^2\rho^j = \frac{\rho(1+\rho)}{(1-\rho)^3},$$

applied to the tail $k > 150$. The resulting interval value is

$$0.002895223101329 \dots < 0.002895224.$$

□

Lemma 34 (Numerical certificate). *For the sequence in Definition 28, (λ_0, m, g) is admissible and*

$$\log D_{\lambda_0, m} < 3.281773, \quad \Omega_{\lambda_0, m, g} < 0.066615.$$

Consequently,

$$\frac{e^\gamma D_{\lambda_0, m}}{(1 - \Omega_{\lambda_0, m, g})^2} < 54.43.$$

Proof. Lemma 30 and Lemma 31 show that $m_k \leq N_k$ for every k .

For $0 \leq k \leq 32$, Lemma 30 gives

$$\sum_{k=0}^{32} \ell_k(m_k) < 3.277487.$$

For $k \geq 33$, every prime in I_k is at least Y_k , so

$$\ell_k(m_k) \leq -m_k \log \left(1 - \frac{1}{Y_k} \right) \leq \frac{m_k}{Y_k - 1}.$$

Lemma 33 gives

$$\sum_{k=33}^{\infty} \ell_k(m_k) < 0.004286.$$

Therefore

$$\log D_{\lambda_0, m} < 3.277487 + 0.004286 = 3.281773.$$

Next, Lemma 30 gives

$$\sum_{k=0}^{32} g_k s_k(m_k) < 0.062611.$$

For $k \geq 33$, Definition 28 gives $m_k = \lfloor \mu_0 Y_k^{2/3} \rfloor$, hence $m_k + 1 > \mu_0 Y_k^{2/3}$. Together with Lemma 32, this gives

$$s_k(m_k) \leq \frac{1.682294}{\sqrt{\mu_0}} Y_k^{-1/3}.$$

By Lemma 33,

$$\sum_{k=33}^{\infty} g_k s_k(m_k) < \frac{1.682294}{\sqrt{1.48035}} 0.002895224 < 0.004004.$$

Thus

$$\Omega_{\lambda_0, m, g} < 0.062611 + 0.004004 = 0.066615 < 1.$$

The preceding bound for $\log D_{\lambda_0, m}$ gives $D_{\lambda_0, m} < \infty$, and $g_k \rightarrow \infty$ is immediate from the definition of g_k . Hence (λ_0, m, g) is admissible. Finally,

$$\frac{e^{\gamma+3.281773}}{(1 - 0.066615)^2} < 54.43.$$

□

Corollary 35 (Optimized weighted asymptotic constant). *If n is large enough, then every n -admissible pair satisfies*

$$|A| |B| < \frac{54.43n^2}{\log n}.$$

Proof. Apply Theorem 27 with (λ, m, g) as in Definition 28, and use Lemma 34. □

REFERENCES

- [1] P. Dusart, *Explicit estimates of some functions over primes*, Ramanujan Journal **45** (2018), 227–251.